

Name 1:

Name 2:

Group #:

<i>Question #</i>	<i>Number of Points</i>
1 - T/F and MC	/10
2 - Variables & Data Types	/10
3 - Lists & Loops	/10
4 - Problem Solving	/15
Total	45

Question #1

Total Points: (10) | 1 mark each

- a) Which of the following variable names is **NOT valid** in Python?
- a. student_name
 - b. _score
 - c. 2ndPlace
 - d. totalMarks
- b) The `input()` function always stores the user's input as a string.
- a. True
 - b. False
- c) Which data type would best store the value `3.75`?
- a. int
 - b. str
 - c. float
 - d. Bool
- d) Which statement is correct regarding file input/output in Python?
- a. `readline()` reads the entire file at once.
 - b. `write()` is used for reading from a file.
 - c. `open()` is only used for creating new files.
 - d. `close()` is not necessary when working with files.
- e) What will be printed by the following code?
- ```
numbers = [2, 4, 6]
print(numbers[1])
```
- a. 2
  - b. 4
  - c. 6
  - d. Error
- f) Which statement correctly describes an `if` statement?
- a. It repeats code a fixed number of times
  - b. It executes code only if a condition is true
  - c. It stores multiple values
  - d. It defines a function

- g) How do you access the value associated with the key 'age' in a dictionary named person?
- a. person['age']
  - b. person.get('age')
  - c. person.value('age')
  - d. person[age]
- h) Which of the following creates an **empty list**?
- a. list = ()
  - b. list = {}
  - c. list = []
  - d. list = ""
- i) What keyword is used to stop a loop immediately?
- a. stop
  - b. break
  - c. exit
  - d. Continue
- j) What will be the output of this pseudocode?
- ```
FOR i FROM 1 TO 10 DO
    IF i MOD 3 = 0 THEN
        PRINT i
    ENDIF
ENDFOR
```
- a. 3 6 9
 - b. 1 2 3 4 5 6 7 8 9 10
 - c. 0 3 6 9
 - d. 3 6 9 12

Question #2

Total Points: (10)

A **library system** is being created to track users and their borrowing activity.

The following code is part of the program:

main.py

UTF-8

```
member_name = "Sam"  
books_borrowed = 3  
has_overdue = False
```

- a) (3 points) Identify the **data type** of each variable and briefly explain **why that data type is appropriate** for its use in this program.
- b) (2 points) Will this line of code run successfully?
- If **yes**, explain what it does
 - If **no**, explain why it causes an error

main.py

UTF-8

```
books_borrowed = books_borrowed + "2"
```

- c) (2 points) What is the value of **total_fee**, and what is its **data type**?

main.py

UTF-8

```
late_fee = 1.5  
days_late = 4  
total_fee = late_fee * days_late
```

- d) (3 points) Rewrite **one** of the original variable names to make it **more descriptive**, and explain your choice.

Question #3

Total Points: (10)

A sports club registration system stores player names and their jersey numbers.

main.py

UTF-8

```
players = ["Aiden", "Bella", "Chris", "Dina"]
jerseys = {"Aiden": 7, "Bella": 12, "Chris": 5}
```

a) (2.5 points) What will be printed by the following code?

main.py

UTF-8

```
print(players[1] + " wears jersey " + str(jerseys["Bella"]))
```

b) (2.5 points) Write one line of code to add "Dina" with jersey number 9 to the jerseys dictionary.

c) (5 points) Explain what the following code does and give its output:

main.py

UTF-8

```
if "Dina" in jerseys:
    print("Jersey assigned")
else:
    print("No jersey found")
```

Question #4

Total Points: (5) + (10) BONUS!!

A movie ticket booking system is being built using Python. Each ticket costs a fixed price, but the total depends on how many tickets are purchased.

- a) (2 points) Explain **why a function is a better choice** than writing the same calculation multiple times in the program.
- b) (3 points) Write a function called `calculate_price(ticket_price, quantity)` that:
 - Multiplies `ticket_price` by `quantity`
 - Returns the total price
 - Works correctly for **any valid quantity**
- c) **Bonus Question** (10 points) Before confirming a booking, the system checks seat availability. Write **pseudocode** that:
 - Asks the user how many seats they want
 - If the number is greater than 5, print `"Not enough seats"`
 - Otherwise, print `"Booking confirmed"`